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Restoration of energy metabolism in leukemic mice treated by a siddha drug—*Semecarpus anacardium* Linn. nut milk extract

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Abstract

Chronic myeloid leukemia (CML) is a clonal disorder characterized by proliferation of hematopoietic cells that possess the *BCR–ABL* fusion gene resulting in the production of a 210 kDa chimeric tyrosine kinase protein. CML, when left untreated, progresses to a blast phase during which the disease turns aggressive and shows poor response to known treatment regimens. We have studied a Siddha herbal agent, *Semecarpus anacardium* Linn. nut milk extract (SA) for its antileukemic activity and its effect on the changes in energy metabolism in leukemic mice. Leukemia was induced in BALB/c mice by tail vein injection of *BCR–ABL*$^+$ 12B1 murine leukemia cell line. This resulted in an aggressive leukemia, similar to CML in blast crisis, myeloid subtype, confirmed by histopathological study and RT-PCR for the p210 mRNA in the peripheral blood, spleen and liver. Leukemia-bearing mice showed a significant increase in lipid peroxides, glycolytic enzymes, a decrease in gluconeogenic enzymes and significant decrease in the activities of TCA cycle and respiratory chain enzymes as compared to control animals. SA treatment was compared with standard drug imatinib mesylate. SA administration to leukemic animals resulted in clearance of the leukemic cells from the bone marrow and internal organs on histopathological examination and this was confirmed by RT-PCR for the p210 mRNA. Treatment with SA significantly reversed the changes seen in the levels of the lipid peroxides, the glycolytic enzymes, the gluconeogenic enzymes and the mitochondrial enzymes. These effects are probably due to the flavonoids, polyphenols and other compounds present in SA which result in total regression of leukemia and correction of the alterations in energy metabolism. Study of animals treated with SA alone did not reveal any adverse effects. On the basis of the observed results, SA can be considered as a readily accessible, promising and novel antileukemic chemotherapeutic agent.
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Keywords: *Semecarpus anacardium*; 12B1 cell line; Murine leukemic model; *BCR–ABL* fusion gene; Glycolytic enzymes; Mitochondrial enzymes

1. Introduction

Chronic myeloid leukemia (CML) is a malignant clonal disorder of hematopoietic stem cells leading to massive expansion of myeloid lineage cells at all stages of maturation. CML is characterized by two distinct clinical phases [1]. The first or chronic phase is marked by a massive increase in leukocyte count in the peripheral

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blood with leukocytes showing a full range of maturation, a marked increase in the myeloid erythroid ratio in the bone marrow and massive splenomegaly. The median survival of patients with CML after diagnosis is 4–5 years. If left untreated, a majority of cases of CML enter the phase of blast crisis, during which the disease turns aggressive and becomes an acute leukemia (of lymphoid, myeloid or mixed type). Once the blast crisis phase is entered, long-term survival is uncommon with most patients expiring within 3–6 months [1].

Development of CML is associated with a specific chromosomal translocation known as the "Philadelphia (Ph) chromosome". The Philadelphia chromosome was the first consistent chromosome abnormality found in any kind of malignancy and was one of the first translocations known to have a direct role in the pathogenesis of a malignant tumor [1]. The Ph chromosome is a shortened chromosome 22 resulting from a reciprocal translocation, *t* (9; 22) (q34; q11), between the long arms of chromosomes 9 and 22 [2,3]. The Ph translocation adds a 3′ segment of the ABL gene from chromosome 9q34 to the 5′ part of the BCR gene on chromosome 22q11, creating a hybrid *BCR–ABL* gene that is transcribed into a chimeric *BCR–ABL* messenger RNA (mRNA). The chimeric mRNA results in the production of a *BCR–ABL* fusion protein with abnormal tyrosine kinase activity. This translocation is strongly implicated in the pathogenesis of chronic myeloid leukemia and a small proportion of acute leukemias (acute lymphoblastic leukemia and acute myeloid leukemia). Introduction of the *BCR–ABL* fusion gene into normal hematopoietic cells of mice causes immortalization of these cells. Injection of such cells into normal mice results in a disorder resembling CML. The 12B1 cell line is a *BCR–ABL* positive cell line, derived by retroviral transformation of bone marrow cells of BALB/c mice with the human *BCR–ABL* fusion gene and expresses the p210 *BCR–ABL* protein [4]. Injection of this cell line into normal BALB/c mice induces an aggressive leukemia resembling CML in blast crisis. The 100% lethal dose (LD100) is 100 cells after tail vein injection [5]. This animal model was first established by He et al. [5] and further characterized by Sobotkova et al. [6] and Jelinek et al. [7].

Derangements in energy metabolism are commonly associated with malignant tumors. Malignant tumors from either humans or animal models lack metabolic flexibility and are largely dependent on glucose for energy. Enhanced glycolysis produces excess lactic acid that can return to the tumor as glucose through the Cori cycle. A high glycolytic rate is important for rapidly proliferating cancers, not only as a major energy source, but also to provide such cells with precursors for nucleotide and lipid biosynthesis. In addition to glycolytic dependence, most tumors express abnormalities in the number and function of their mitochondria. The principal mitochondrial substrate is pyruvate formed essentially by glycolysis. Pyruvate carbons enter the tricarboxylic acid cycle (TCA cycle) in the mitochondrial matrix, in which substrates are oxidized with the formation of CO_2 and the reduction of NAD^+ and FAD^+ to NADH and $FADH_2$, respectively. These provide electrons to the respiratory chain upon their re-oxidation. Electron transfer from the TCA cycle to the respiratory chain by NADH and $FADH_2$ promotes the generation of ATP. It has been established that defects in the respiratory chain enzymes lead to enhanced production of ROS and free radicals in mitochondria, resulting in mitochondrial DNA mutations which indirectly impair glucose sensing by reducing intracellular concentrations of ATP [8], an important metabolic fuel.

A number of phytochemicals derived from plants possess novel structures with anticancer potential—this makes them interesting tools in cancer research, both in experimental animal systems and in humans [9]. *Semecarpus anacardium* (Anacardiaceae) Linn. commonly known as 'marking nut', has found many applications in Indian medicine, in the treatment of cancer, rheumatoid arthritis and gout [10]. Acetylated oil of *S. anacardium* potentiated the efficacy of widely used anticancer drugs such as mitomycin, fluorouracil and methotrexate [11]. It has also been reported that Ayurvedic oil preparation from SA nut extract has cytotoxic activity on the leukemic cell lines K562 and HL 60 [12]. The nut possesses a spectrum of bioflavanoids, such as semecarpoflavonone [13], jeediflavonone [14], galluflavonone [15], nallaflavonone [16], semecarpetin [17] and anacardioflavonone [18] along with phenolic compounds, bhilawanols, sterols and glycosides The phytochemical studies of the milk extract have been shown in our laboratory to contain flavonoids, phenols, and carbohydrates [19] was subjected to investigation against hepatocellular carcinoma [10], mammary carcinoma [20,21] and rheumatoid arthritis [22] in experimental animals. Numerous studies have demonstrated the protective properties of polyphenolic flavonoids. Flavonoids as effective antioxidants may provide protection against cancer [23]. Also *in vitro* they were found to be antiproliferative against broad spectrum of cancer cells [24]. The pharmacological active principles of this plant are flavonoids. The ethyl acetate extract contains a biflavonoid known as tetrahydroamentoflavone (THA) that inhibited the enzyme cyclooxygenase-2 (COX-2) [25]. The alcoholic extract of dry nuts had anti-fungal activity while of nut shells were shown to prevent lipid peroxidation [26]. The nut milk extract is effective in restoring the fragility of lysosomal membranes in aflatoxin-induced hepatocellular carcinoma [27].

Hence the current study is aimed at evaluating the antileukemic effect of *S. anacardium* Linn. nut milk extract (SA) as well as its effect on carbohydrate metabolizing, mitochondrial TCA cycle and respiratory chain enzymes in a mouse model with aggressive *BCR–ABL*$^+$ leukemia, analogous to CML in blast crisis.

2. Materials and methods

BCR–ABL$^+$ leukemia was induced in 6–10-week-old female BALB/c mice by a single tail vein injection of the 12B1 cell line.

2.1. Animals and diet

Female BALB/c mice (6–10 weeks old) were obtained from Tamil Nadu Veterinary and Animal Sciences University Laboratories, Chennai, India. The animals were housed in spacious hygienic plastic cages in an air-conditioned room controlled for temperature (27 ± 1 °C), humidity (60 ± 5%), lighting 12-h (light–dark cycle). The mice were fed a commercial mice pelleted diet (Gold Mohur mice feed, Hindustan lever Limited, Mumbai, India) and water *ad libitum*. Experimental animals were handled according to the University and Institutional Legislation, regulated by the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), Ministry of Social Justice and Empowerment, Government of India (IAEC no. 06/003/04).

2.2. Drugs and chemicals

S. anacardium Linn. nut milk extract, a Siddha formulation was prepared according to a recipe of the 'Formulary of Siddha Medicine' [28]. RPMI, fetal calf serum, glutamine, penicillin and streptomycin sulfate, amphotericin B, modified Eagle's medium non-essential amino acids, pyruvate, and β-mercaptoethanol all from Gibco/BRL. All other chemicals and solvents used were of analytical grade.

2.3. BCR–ABL positive leukemia cell line for induction

12B1 is a murine leukemia cell line derived by retroviral transformation of BALB/c bone marrow cells with the human *BCR–ABL* (b_3 a_2) fusion gene and expresses the p210 *BCR–ABL* protein [4]. This induces an aggressive leukemia in BALB/c mice, the 100% lethal dose being 100 cells after tail vein injection [5]. The 12B1 cell line was a kind gift from Dr. Emmanuel Katsanis (Department of Pediatrics, Steele Memorial Children's Research Center, University of Arizona, Tucson, AZ, USA). 12B1 cells were cultured at 37 °C under 5% CO_2 in RPMI1640 medium containing 10% heat-inactivated fetal calf serum and supplemented with 2 mM glutamine, 100 U/ml penicillin, 100 μ/ml streptomycin sulfate, 0.025 μg/ml amphotericin B, 0.1 mM modified Eagle's medium non-essential amino acids, 1 mM pyruvate, and 50 μM 2-mercaptoethanol.

The cell lines were tested monthly and were always found to be free of *Mycoplasma* contamination. *Mycoplasma* species identification by colorimetric assay was performed with commercial diagnostic Mycoplasma detection kit (PlasmoTest^{TM} Kit, InvivoGen, Lonza Group Ltd., Switzerland) following the manufacturer's instructions. The kit detects all *Mycoplasma* and *Acheloplasma* species known to infect cell cultures, as well as other cell culture contaminants such as bacteria.

In vitro experiments were carried out to determine the effect of SA on 12B1 cells. The results showed a marked dose-dependent reduction in cell survival. There was 60–70% inhibition of cell proliferation upon exposure to 60 μg/ml (fixed as optimum concentration) of SA. Based on the time kinetics, there was no difference between 24 and 48 h incubations. It was inferred that the incubation of SA had led to marked decrease in the number of viable 12B1 *Bcr–Abl*$^+$ murine leukemic cells (communicated).

2.4. Experimental design

The animals were randomly divided into five groups of six animals each. Group I served as control mice receiving olive oil, a vehicle, alone. Group II mice were induced by a single tail vein injection of 12B1 cell line (100 cells) [5]. After 20 days, induction of leukemia was confirmed by cytological study of bone marrow and histological examination of spleen and liver as well as RT-PCR of peripheral blood, spleen and liver, [5]. Groups III received a standard drug imatinib mesylate (300 mg/kg body weight) twice a day in 300 μl of sterile water orally by gastric incubation for 7 days [29]. Groups IV received SA (200 mg/kg body weight/day) in olive oil orally by gastric intubation for 14 days [10,20,22]. Group V received SA as drug control.

At the end of the experimental period, all animals were killed by cervical decapitation. The heparinized blood was collected and used for total RNA isolation. Smears were made from peripheral blood and bone mar-

Table 1
PCR primers used in this study

BCR–ABL mRNA (385 bp)	
Sense primer: 5′-ACAGAATTCGCTCACCATCAATAAG-3′	
Anti-sense primer: 5′- ACAGAATTCGCTCACCATCAATAAG-3′	
RAR-α mRNA (163 bp)	
Sense primer: 5′-TCCCCAGCCACCATTGAGACC-3′	
Anti-sense primer: 5′-AGCCCTTGCAGCCCTCACAG-3′	

row and stained with Leishman's stain. One smear of bone marrow was stained with the PAS stain and another with Sudan Black B stain as per standard protocols [30]. Necropsy was done and the internal organs were examined. Organs (spleen, liver) were immediately excised from the animals and weighed—one bit was fixed in 10% neutral buffered formalin, routinely processed and embedded in paraffin wax. Tissue sections were subsequently cut and stained with hematoxylin and eosin for histopathological examination [31]. The remaining portions of the spleen and liver were then used for various assays as described below.

2.5. Detection of mRNA expression using reverse transcription-PCR

Reverse transcription (RT)-PCR was used to detect the expression of the *BCR–ABL* fusion gene (b_3a_2). Total RNA was extracted using TRIZOL reagent from blood, spleen and liver of all Groups. First strand cDNA was synthesized using 5 μg total RNA by RT-PCR kit (Genei, Bangalore, India). The related PCR primers listed in Table 1 were used to produce the respective correlated products.

The PCR for *BCR–ABL* and *RAR-α* cDNAs (internal control) were performed with 30 amplification cycles and the reaction conditions were: denaturation for 60 s at 94 °C, primer annealing for 90 s at 55 °C and extension for 90 s at 72 °C, then the cycles were repeated 27 times. After final extension of 5 min at 72 °C, then the reaction was terminated by holding the temperature at 4 °C. The above PCR components were incubated in an Eppendorf mastercycler gradient (Eppendorf, Westbury, N.Y.). The PCR products were then run on 2% agarose gel and visualized by ethidium bromide staining under UV illuminator.

2.6. Isolation of mitochondria

Mitochondrial fraction of spenic and hepatic tissue was isolated according to an established method [32] with some modifications. Briefly, 10% spleen and liver homogenate was prepared in ice-cold 0.25 M sucrose solution. The homogenate was centrifuged at 1000 × *g* for 10 min to obtain the nuclear pellet. Mitochondrial pellet was obtained by centrifuging the post nuclear supernatant at 10,000 × *g* for 20 min. Mitochondrial pellet was finally suspended in 0.25 M sucrose solution. All operations were performed at 4 °C. The mitochondrial fraction was used for TCA cycle and respiratory chain enzyme studies. The purity of the mitochondria was assessed by estimating succinate dehydrogenase activity [33].

2.7. Biochemical assays

The spleen and liver was then washed well with ice-cold isotonic saline and blotted individually on ash-free filter paper. Ten percent spleen and liver homogenate was prepared in Tris–HCl buffer (0.1 M, pH 7.4) using Potter–Elvehjem homogenizer with Teflon pestle. The protein content was determined by the method of Lowry et al. [34]. Tissue lipid peroxides were measured by the method of Devasagayam and Tarachand [35] using malonodialdehyde as standard. The values are expressed in terms of nmoles of MDA formed/min/mg protein.

Hexokinase (HX), phosphoglucoisomerase (PGI) were assayed by the method of Banstrup et al. [36] and aldolase by King [37]. The activity of glucose 6-phosphatase was measured according to the method of King [38] and fructose 1,6-bisphosphatase was assayed by Gancedo and Gancedo [39].

The activity of isocitrate dehydrogenase was assayed by the method of Bernt and Bergmeyer [40]. α-Ketoglutarate activity was assayed according to the colorimetric determination of ferrocyanide produced by the decarboxylation of α-ketoglutarate with ferricyanide as electron acceptor [41]. Succinate dehydrogenase activity was assayed by the method of Slater and Bonner [33] in which the rate of reduction of potassium ferricyanide was measured in the presence of potassium cyanide. The activity of malate dehydrogenase was estimated by the method of Mehler et al. [42]. The method of Minakami et al. [43] was followed for the determination of reduced nicotinamide dinucleotide (NADH) dehydrogenase activity. The activity of cytochrome *c* oxidase was assayed by the method of Wharton and Tzagoloff [44].

2.8. Transmission electron microscope

For electron microscopy, splenic tissue from leukemia-induced and SA-treated animals were fixed in 10% glutaraldehyde, post-fixed in 1% osmium tetroxide in 0.1 M phosphate buffer and embedded in Maraglass 655 (Polysciences Inc., Warrington, PA). Thin sections were stained with 1% Azur II plus 1% methylene blue in water for 3–5 min, then examined and photographed with a light microscope. Ultramicrotome sections, contrasted with uranyl acetate and lead citrate, were examined and photographed with a Zeiss EM-9A transmission electron microscope.

2.9. Statistical analysis

The results obtained were expressed as the mean ± S.D. Student's *t*-test was employed for statistical evaluations. *p*-values <0.05 were considered significant.

3. Results

3.1. SA on body weight

Table 2 represents the body weight changes of the control and experimental animals. The initial and final body weights were noted. Initially, there were no significant differences in the body weight of the control and experimental animals. But eventually, there was a significant decrease in the body weight of the leukemia-bearing (Group II) animals ($p < 0.05$) when compared to control (Group I) animals. Imatinib mesylate-treated (Group III) animals showed a significant increase in body weight ($p < 0.05$) when compared to Group II animals. On treatment with SA (Group IV), the body weights were significantly ($p < 0.05$) increased when compared with Group II animals and moderately increased in weight when compared with Group III animals. Drug control (Group V) animals showed an increase in their body weight but not to significant levels when compared to normal control (Group I) animals.

Necropsy revealed massive enlargement of the spleen and marked enlargement of the liver in Group II (leukemia-induced) animals as compared to the control (Group I), imatinib-treated (Group III), SA-treated (Group IV) and drug control (Group V) Groups.

3.2. Peripheral blood smear examination

The control group of animals (Group I) showed normal RBCs, normal WBCs and normal platelets. Peripheral blood smear from leukemic mice (Group II) revealed normal red blood cells, normal white blood cells with occasional cells resembling myeloblasts, occasional immature myeloid cells like myelocytes and aggregates of platelets. The RBCs, WBCs and platelets appeared normal in the other three Groups (Groups III, Group IV and Group V).

3.3. Changes in bone marrow

Fig. 1a–c shows the bone marrow of control and experimental animals. The bone marrow smear of control (Group I) (Fig. 1a) animals showed normal cellularity with normal erythroid and myeloid series. Scattered normal megakaryocytes were seen. The *BCR–ABL*$^{+}$ leukemia-induced animals (Group II) (Fig. 1b) showed marked hypercellularity with numerous neoplastic cells comprising of blasts which were large and had round to indented nuclei, prominent nucleoli and moderate cytoplasm. The chromatin structure was fine. The normal myeloid, erythroid and megakaryocytic cells were markedly reduced. On treatment with imatinib mesylate (Group III), the bone marrow showed normal erythroid and myeloid and megakaryocytic cells. Upon treatment with SA (Group IV) (Fig. 1c), the bone marrow changes reverted back to normal. Bone marrow smears of drug control animals (Group V) appeared normal, indicating that SA did not produce any adverse effects on hematopoietic cells.

3.4. Histopathological study

3.4.1. Changes in spleen

Fig. 2a–d show the spleen of control and experimental animals. The spleen of control (Group I) (Fig. 2a) animals showed normal histology with a connective cap-

Table 2
Effect of *Semecarpus anacardium* on the body weight changes in control and experimental animals

Parameters	Group I (control)	Group II (leukemia-induced)	Group III (leukemia + imatinib mesylate)	Group IV (leukemia + SA-treated)	Group V (drug control – SA alone)
Body weight					
Initial	17.1 ± 1.1	17.3 ± 1.3	17.3 ± 1.4	17.4 ± 1.6	17.5 ± 1.7
Final	17.8 ± 1.1	12.4 ± 1.0 ^{a*}	16.1 ± 1.2 ^{b*}	16.4 ± 0.9 ^{b*cNS}	18.3 ± 1.1^{dNS}

Values are expressed as mean ± S.D. for six animals. [a] Denotes Group I compared with Group II. [b] Denotes Group II compared with Group III and Group IV. [c] Denotes Group III compared with Group IV. [d] Denotes Group I compared with Group V. Statistical differences are expressed as: $^{*}p < 0.05$. NS, not significant.

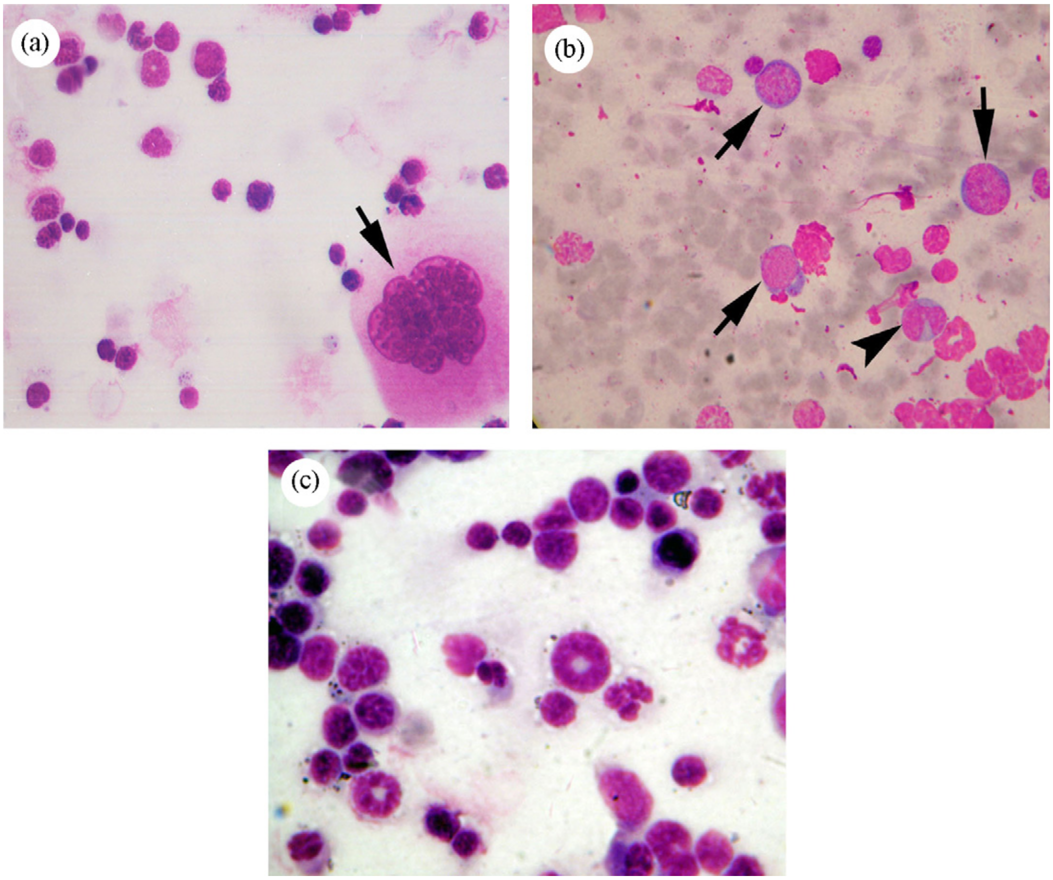

Fig. 1. (a–c) Cytological changes in bone marrow. (a) Bone marrow smear from control mouse (Group I). Normal bone marrow cells including a normal megakaryocyte (arrow) seen (Leishman ×1000). (b) Bone marrow smear from mouse leukemic model (Group II) – shows numerous blasts (arrow) with scanty basophilic cytoplasm, large nuclei, fine nuclear chromatin and nucleoli. One of the blasts shows an indented nucleus (arrow head) (Leishman ×400). (c) Bone marrow smear from leukemic mouse treated with *Semecarpus anacardium* Linn. nut milk extract (Group IV) – shows normal cells with absence of infiltrating blasts (Leisman ×1000).

sule and underlying splenic parenchyma composed of normal red pulp and white pulp. *BCR–ABL*$^{+}$ leukemia-induced animals (Group II) (Fig. 2b,c) showed an effacement of architecture of the spleen with expansion of red pulp, There was prominent extramedullary haematopoiesis with infiltration by myeloid cells and innumerable megakaryocytes as well as infiltration by neoplastic cells with large nuclei and scanty cytoplasm. On administration with imatinib mesylate (Group III), the spleen showed normal histology. Similarly, treatment with SA (Group IV) also restored the architecture and histology of the spleen to normal (Fig. 2d). In drug control animals (Group V), the spleen showed no abnormality.

3.4.2. Changes in liver

In the control mice, the liver was composed of cords of hepatocytes with sinusoids in between. The sinusoids were lined by Kupfer cells. Normal portal tracts and central veins were seen. (Group I) (Fig. 3a). In *BCR–ABL*$^{+}$ leukemia-induced animals (Group II) the liver showed multiple, small foci of extramedullary hematopoiesis and infiltration by neoplastic cells (Fig. 3b, c). On treatment with imatinib mesylate (Group III), the liver showed normal histology. After treatment with SA (Group IV) (Fig. 3d), the histology of liver reverted back to normal with no evidence of any infiltrate. In drug control animals (Group V), the liver showed normal histology.

3.4.3. mRNA expression of *BCR–ABL*$^{+}$ gene by RT-PCR

Fig. 4 depicts the mRNA expression of *BCR–ABL* fusion gene (b_3a_2) as 385 bp and the internal control, *RAR-α* as 163 bp. Lane 1 and 17 (M) show 100 bp DNA ladder. Lanes 2–4 (GI) show the mRNA expression of *RAR-α* (internal control) in control animals

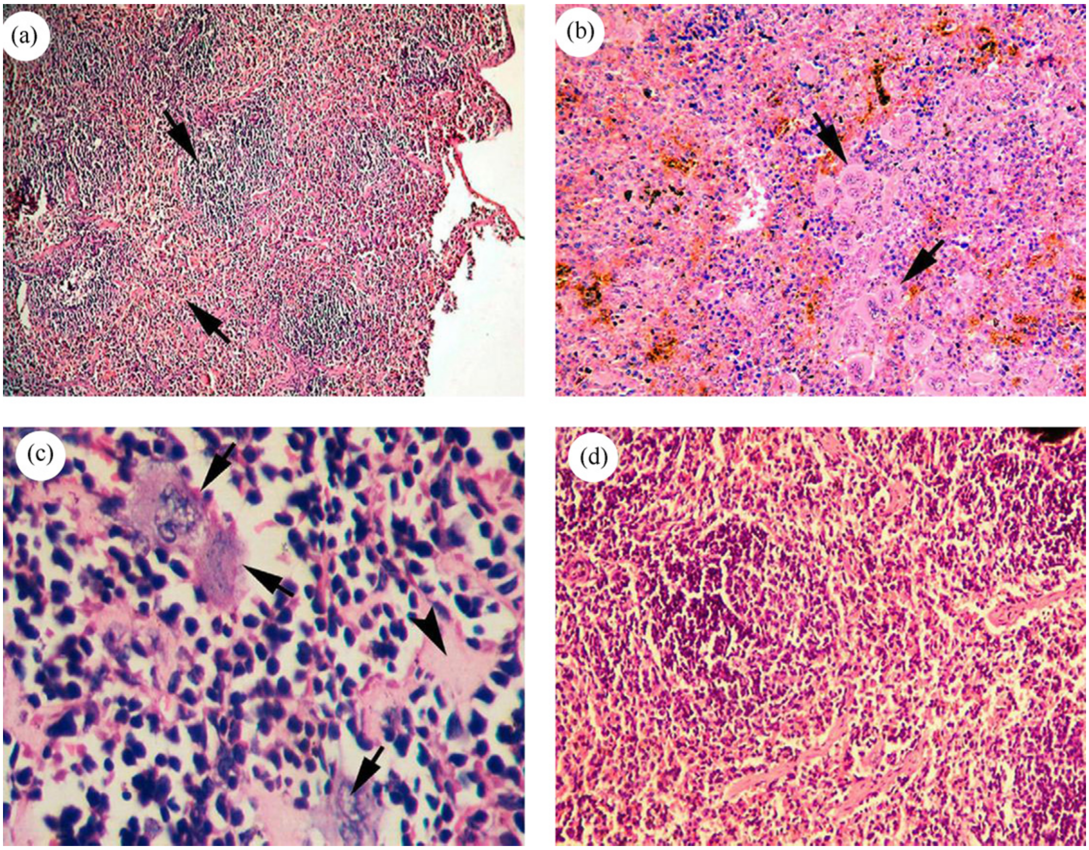

Fig. 2. (a–d) Histological changes in spleen. (a) Section of spleen from control animal (Group I) – showing normal red and white pulp (arrows) (H & E ×100). (b) Spleen from Group II leukemia-induced mouse (Group II) – shows effacement of architecture, infiltration by leukemic cells and extramedullary hematopoiesis with numerous megakaryocytes (arrow) (H & E ×200). (c) Spleen from another animal in (Group II) – showing complete replacement by leukemic cells with dark staining large nuclei and scanty cytoplasm (arrow)-few megakaryocytes seen (arrows). Small foci of hyalinization are present (arrowhead) (H & E ×400). (d) Section of spleen from leukemia-induced mouse treated with *Semecarpus anacardium* Linn. nut milk extract (Group IV) – shows reversion to normal architecture with absence of extramedullary hematopoiesis and absence of leukemic infiltrate (H & E ×400).

(Group I). Lane 5–7 (GII) show the mRNA expression of *BCR–ABL* fusion gene (385 bp) in peripheral blood, spleen and liver of leukemia-induced mice. Lanes 8–10 (GIII) depict the mRNA expression of *BCR–ABL* in imatinib mesylate-treated animals (Group III). The blood and liver of this group showed complete suppression of *BCR–ABL* fusion gene and the spleen showed markedly decreased levels. Lanes 11–13 show the mRNA expression of *BCR–ABL* fusion gene of SA-treated (Group IV) animals. The peripheral blood and liver of Group IV animals shows the complete absence of *BCR–ABL* fusion gene while the spleen shows a faintly visible band. SA-treated animals and imatinib mesylate-treated animals show similar expression. Drug control animals (lanes 14–16-GV) show no marked changes in their mRNA expression.

3.4.4. SA on survival of leukemic mice

Fig. 5 depicts the survival of treated and untreated animals. The leukemia-induced animals survived 20–28 days whereas the leukemic mice, after SA treatment survived for 490–510 days.

3.4.5. SA on lipid peroxides

Table 3 shows the effect of treatment with SA on the frequencies of leukemia-induced lipid peroxides with inducers ($FeSO_4$, ascorbate and H_2O_2) in the spleen and liver. Even in the absence of any inducer, the basal levels of lipid peroxides, were found to be significantly (spleen and liver $p < 0.05$) increased in the leukemic animals when compared to controls. In the presence of inducers, $FeSO_4$, ascorbate and H_2O_2, the levels of lipid peroxides were found to be significantly (spleen and liver $p < 0.05$)

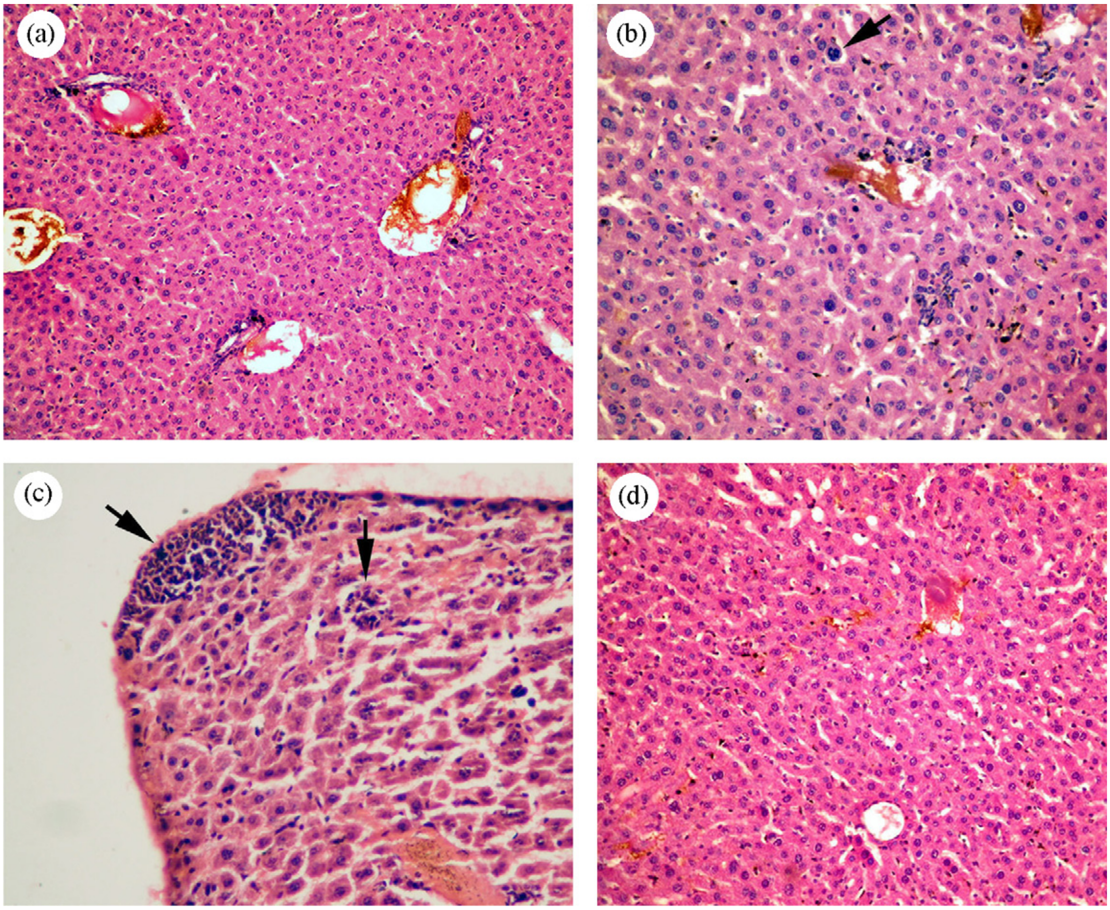

Fig. 3. (a–c) Histological changes in liver. (a) Liver from control mouse (Group I) – normal hepatocytes and portal tracts (H & E ×200). (b) Liver from leukemia-induced mouse (Group II) – shows extramedullary hematopoiesis – a megakaryocyte is seen (arrow) (H & E ×400). (c) Liver from leukemic mouse treated with *Semecarpus anacardium* Linn. nut milk extract (Group IV) showing disappearance of infiltrates (H & E ×200).

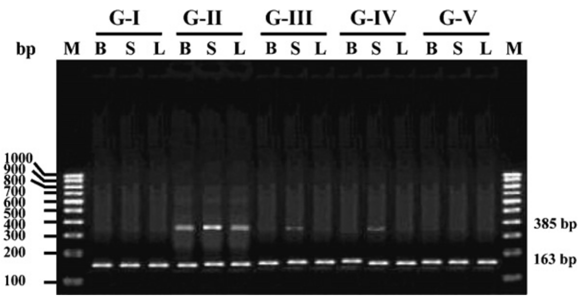

Fig. 4. mRNA expression of the *Bcr–Abl* fusion gene (b3a2) in peripheral blood, spleen and liver BALB/c leukemic mice treated with SA by RT-PCR amplification. M, marker; S, spleen; L, liver; B, blood. G-I, control animals (Group I); G-II, *Bcr–Abl* leukemia-induced animals (Group II); G-III, imatinib mesylate-treated animals (Group III); G-IV, SA-treated animals (Group IV); G-V, drug control animals (Group V).

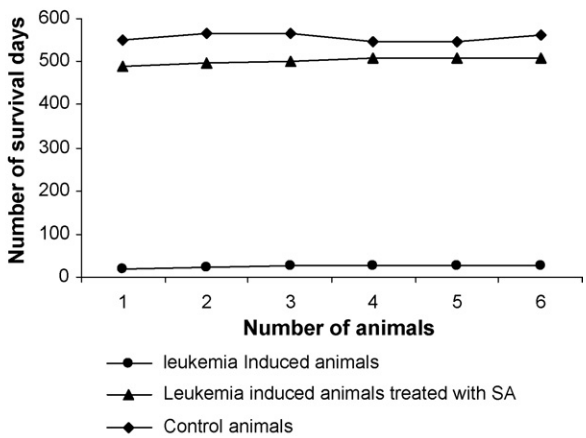

Fig. 5. Survival curves of control mice, leukemia-induced mice (injected with 12B1 leukemic cells) and leukemia-induced (injected with 12B1 leukemic cells) mice treated with SA.

Table 3
Effect of *Semecarpus anacardium* Linn. nut milk extract on lipid peroxides spleen and liver of control and experimental animals

Parameters (nmoles of MDA formed/min/mg protein	Group I (control)	Group II (leukemia-induced)	Group III (leukemia + imatinib mesylate	Group IV (leukemia + SA-treated)	Group V (drug control – SA alone)
Spleen					
Basal	0.79 ± 0.23	1.86 ± 0.12^{a*}	0.93 ± 0.09^{b*}	1.09 ± 0.09^{b*c*}	0.76 ± 0.24 ^{dNS}
Ferrous sulphate	6.28 ± 0.32	9.49 ± 0.15 ^{a*}	6.81 ± 0.89 ^{b*}	7.41 ± 0.89 ^{b*c*}	6.27 ± 0.32 ^{dNS}
Ascorbate	2.79 ± 0.24	4.86 ± 0.21 ^{a*}	3.26 ± 0.38 ^{b*}	4.06 ± 0.38 ^{b*c*}	2.77 ± 0.23 ^{dNS}
Hydrogen peroxide	1.84 ± 0.19	3.08 ± 0. 12 ^{a*}	2.03 ± 0.21 ^{b*}	2.23 ± 0.18 ^{b*cNS}	1.86 ± 0.19 ^{dNS}
Liver					
Basal	1.14 ± 0.19	3.28 ± 0.14 ^{a*}	1.46 ± 0.19 ^{b*}	1.96 ± 0.19 ^{b*c*}	1.16 ± 0.17 ^{dNS}
Ferrous sulphate	7.29 ± 0.21	12.69 ± 1.36 ^{a*}	8.24 ± 0.12 ^{b*}	9.64 ± 0.12 ^{b*cNS}	7.31 ± 0.24 ^{dNS}
Ascorbate	4.18 ± 0.07	7.78 ± 0.17 ^{a*}	4.71 ± 0.56 ^{b*}	5.81 ± 0.56 ^{b*cNS}	4.19 ± 0.09 ^{dNS}
Hydrogen peroxide	1.49 ± 0.19	2.76 ± 0.16 ^{a*}	1.71 ± 0.17 ^{b*}	2.11 ± 0.17 ^{b*cNS}	1.48 ± 0.19 ^{dNS}

Values are expressed as mean ± S.D. for six animals. Comparisons are made between: a, Group I and Group II; b, Group II and Group III, Group IV; c, Group III and Group IV; d, Group I and Group V. Statistical significance are expressed as: $^{*}p < 0.05$ and NS, not significant.

Table 4
Effect of *Semecarpus anacardium* Linn. nut milk extract on glycolytic enzyme activities on spleen and liver of control and experimental animals

Enzyme (mg protein^{-1})	Group I (control)	Group II (leukemia-induced)	Group III (leukemia + imatinib mesylate)	Group IV (leukemia + SA-treated)	Group V (drug control – SA alone)
Spleen					
Hexokinase	11.67 ± 1.12	18.71 ± 1.47^{a*}	12.98 ± 1.21^{b*}	14.92 ± 1.29^{b*cNS}	10.21 ± 0.98^{dNS}
Aldolase	25.24 ± 2.26	31.62 ± 2.61^{a*}	26.01 ± 2.16^{b*}	28.84 ± 2.32^{b*c*}	28.71 ± 2.04^{dNS}
Phosphoglucoisomerase	20.25 ± 1.87	29.47 ± 2.26^{a*}	20.96 ± 1.78^{b*}	22.73 ± 2.23^{b*cNS}	20.19 ± 1.92^{dNS}
Liver					
Hexokinase	14.81 ± 1.36	28.54 ± 2.43^{a*}	15.08 ± 1.26^{b*}	17.63 ± 1.28^{b*cNS}	14.76 ± 1.24^{dNS}
Aldolase	21.96 ± 2.21	31.40 ± 2.78^{a*}	22.69 ± 2.16^{b*}	25.62 ± 2.25^{b*cNS}	22.07 ± 2.07^{dNS}
Phosphoglucoisomerase	24.82 ± 2.15	34.65 ± 3.21^{a*}	25.28 ± 2.51^{b*}	26.33 ± 2.37^{b*cNS}	24.96 ± 2.19^{dNS}

Values are expressed as mean ± S.D. for six animals. One unit of enzyme activity was expressed as: hexokinase, nmol glucose-6-phosphate; phosphoglucoisomerse, nmol fructose; aldolase, nmol glyceraldehyde formed min^{-1} (mg protein)$^{-1}$ at 37 °C. [a] Denotes Group I compared with Group II. [b] Denotes Group II compared with Group III and Group IV. [c] Denotes Group III compared with Group IV. [d] Denotes Group I compared with Group V. Statistical differences are expressed as: $^{*}p < 0.05$. NS, not significant.

increased in leukemia-bearing (Group II) animals. On treatment with imatinib mesylate (Group III) and SA (Group IV) the lipid peroxide levels, basal and induced (spleen and liver $p < 0.05$) were found to be significantly decreased when compared with untreated (Group II) animals and similar to that of that of the control group. Drug control (Group V) animals did not show any significant changes when compared with control (Group I) animals.

3.4.6. SA on energy metabolism

Table 4 depicts the effect of SA on glycolysis (hexokinase, aldolase and phosphoglucoisomerase) in spleen and liver of control and experimental animals. The activities of hexokinase, aldolase and phosphoglucoisomerase were significantly increased ($p < 0.05$) in *BCR–ABL*$^+$ leukemia-bearing (Group II) animals as compared to control (Group I) animals. These changes were reverted back to near normal levels on imatinib mesylate (Group III) and SA treatment in Group IV animals and the decrease was statistically significant ($p < 0.05$). Drug control mice showed no difference when compared with the controls.

Table 5 shows the activities of the gluconeogenic enzymes (glucose 6-phosphatase and fructose 1,6 bisphosphatase) in spleen and liver of control and experimental animals. The activity of the gluconeogenic enzymes (glucose 6-phosphatase and fructose 1,6 bisphosphatase) were significantly ($p < 0.05$) decreased in Group II when compared to Group I animals. Upon treatment with imatinib mesylate (Group III) the gluconeogenic enzymes showed significantly ($p < 0.05$) increased activity when compared with *BCR–ABL*$^+$ leukemia-bearing animals (Group II). On administration of SA (group IV) the spleen and liver shows significantly increased ($p < 0.05$) activities of glucose 6-phosphatase and fructose 1,6 bisphosphatase when compared to Group II animals. Imatinib mesylate-treated animals (Group III) depict significantly increased ($p < 0.05$)

Table 5
Effect of *Semecarpus anacardium* Linn. nut milk extract on gluconeogenic enzyme activity on spleen and liver of control and experimental animals

Enzyme (units mg/protein)	Group I (control)	Group II (leukemia-induced)	Group III (leukemia + imatinib mesylate)	Group IV (leukemia + SA-treated)	Group V (drug control – SA alone)
Spleen					
Glucose-6-phosphatase	29.91 ± 2.37	17.64 ± 1.87^{a*}	28.16 ± 2.16^{b*}	26.01 ± 2.29^{b*cNS}	30.43 ± 2.57^{dNS}
Fructose 1,6-diphosphatase	35.57 ± 3.08	25.69 ± 2.04 ^{a*}	33.69 ± 2.96 ^{b*}	31.15 ± 2.73^{b*c*}	35.68 ± 3.01^{dNS}
Liver					
Glucose-6-phosphatase	21.60 ± 1.62	10.72 ± 1.16 ^{a*}	20.96 ± 1.56 ^{b*}	17.84 ± 1.56^{b*c*}	21.81 ± 1.68^{dNS}
Fructose 1,6-diphosphatase	30.96 ± 2.51	19.89 ± 1.29 ^{a*}	29.01 ± 2.15 ^{b*}	27.92 ± 2.25^{b*cNS}	31.40 ± 2.47^{dNS}

Values are expressed as mean ± S.D. for six animals. One unit of enzyme activity was expressed as: glucose-6-phosphatase and fructose 1,6-diphosphatase, nmol inorganic phosphorus released min^{-1} (mg protein)$^{-1}$ at 37 °C. ^{a} Denotes Group I compared with Group II. ^{b} Denotes Group II compared with Group III and Group IV. ^{c} Denotes Group III compared with Group IV. ^{d} Denotes Group I compared with Group V. Statistical differences are expressed as $^{*}p < 0.05$. NS, not significant.

activities of fructose 1,6 bisphosphatase of spleen and liver when compared with SA-treated (Group IV) animals. Drug control animals (Group V) did not show any significant changes when compared to control animals (Group I).

The activities of TCA cycle enzymes (isocitrate dehydrogenase, α-ketoglutarate dehydrogenase, succinate dehydrogenase and malate dehydrogenase) and respiratory chain enzymes (NADH dehydrogenase and cytochrome *c* oxidase) in the mitochondrial fractions of spleen and liver of control and experimental animals are shown in Tables 6 and 7, respectively. The activity of the mitochondrial enzymes (isocitrate dehydrogenase, α-ketoglutarate dehydrogenase, succinate dehydrogenase and malate dehydrogenase) and respiratory chain enzymes (NADH dehydrogenase and cytochrome C oxidase) were significantly decreased in *BCR–ABL*$^{+}$ leukemia-bearing animals (Group II) ($p < 0.05$) when compared to control (Group I) animals. On treatment with imatinib mesylate, the tricarboxylic acid (TCA) cycle enzymes and respiratory chain enzymes showed a significantly increased activity ($p < 0.05$) when compared with *BCR–ABL*$^{+}$ leukemia-bearing animals (Group II). Upon SA administration, the TCA cycle enzymes and respiratory chain enzymes showed a significantly increased activity ($p < 0.05$) when compared to *BCR–ABL*$^{+}$ leukemia-bearing animals.

The spleen of imatinib mesylate-treated animals (Group IV) showed a significantly increased activity of isocitrate dehydrogenase, α-ketoglutarate dehydrogenase, malate dehydrogenase and NADH dehydrogenase ($p < 0.05$) when compared to SA-treated (Group III) animals. Liver of imatinib mesylate-treated animals (Group III) showed a significantly increased activity in isocitrate dehydrogenase α-ketoglutarate dehydrogenase and malate dehydrogenase ($p < 0.05$) when compared with SA-treated (Group IV) animals. Drug control animals (Group V) showed no significant variation in TCA cycle enzymes and respiratory chain enzymes when compared to control animals (Group I).

Table 6
Effect of *Semecarpus anacardium* Linn. nut milk extract on the TCA cycle and respiratory chain enzyme activities on spleen of control and experimental animals

Parameters	Group I (Control)	Group II (leukemia-induced)	Group III (leukemia + imatinib mesylate	Group IV (leukemia + SA-treated)	Group V (drug control – SA alone)
Isocitrate dehydrogenase	861.00 ± 49.0	621.00 ± 36.28^{a*}	849.00 ± 47.00^{b*}	831.00 ± 40.12^{b*c*}	861.00 ± 41.21^{dNS}
α-Ketoglutarate dehydrogenase	43.93 ± 4.38	30.81 ± 3.29 ^{a*}	41.86 ± 4.38 ^{b*}	38.73 ± 3.81 ^{b*c*}	43.90 ± 3.71^{dNS}
Succinate dehydrogenase	9.86 ± 0.98	3.89 ± 0.49 ^{a*}	9.01 ± 0.86 ^{b*}	7.21 ± 0.71 ^{b*cNS}	9.96 ± 0.51^{dNS}
Malate dehydrogenase	214.24 ± 19.61	128.17 ± 12.89 ^{a*}	206.24 ± 17.81 ^{b*}	190.28 ± 16.12 ^{b*c*}	215.21 ± 15.11^{dNS}
NADH dehydrogenase	18.01 ± 1.71	9.58 ± 0.71 ^{a*}	17.11 ± 1.61 ^{b*}	14.97 ± 1.41 ^{b*c*}	18.61 ± 1.01^{dNS}
Cytochrome C oxidase	4.11 ± 0.38	2.01 ± 0.42 ^{a*}	3.75 ± 0.42 ^{b*}	2.91 ± 0.51 ^{b*cNS}	4.12 ± 0.47^{dNS}

Values are expressed as mean ± S.D. for six animals. Isocitrate dehydrogenase, nmol of α-ketoglutarate liberated/(min mg protein). α-Ketoglutarate dehydrogenase, μmol of ferricyanide liberated/(min mg protein); succinate dehydrogenase, μmol of succinate oxidized (min mg protein); malate dehydrogenase, nmol of NADH oxidized (min mg protein); NADH dehydrogenase, μmol NADH oxidized (min mg protein). Cytochrome C oxidase, optical density × 10^{-2} min (mg protein)$^{-1}$. ^{a} Denotes Group I compared with Group II. ^{b} Denotes Group II compared with Group III and Group IV. ^{c} Denotes Group III compared with Group IV. ^{d} Denotes Group I compared with Group V. Statistical differences are expressed as $^{*}p < 0.05$. NS, not significant.

Table 7
Effect of *Semecarpus anacardium* Linn. nut milk extract on the TCA cycle and respiratory chain enzyme activity on liver of control and experimental animals

Parameters	Group I (control)	Group II (leukemia-induced)	Group III (leukemia + imatinib mesylate	Group IV (leukemia + SA-treated)	Group V (drug control − SA alone)
Isocitrate dehydrogenase	847.00 ± 46.00	540.16 ± 30.00^{a*}	826.10 ± 38.00^{b*}	798.00 ± 41.00^{b*c*}	847.61 ± 40.00^{dNS}
α-Ketoglutarate dehydrogenase	96.97 ± 8.19	64.25 ± 6.17 ^{a*}	94.27 ± 7.01^{b*}	90.40 ± 8.16 ^{b*c*}	97.79 ± 8.16^{dNS}
Succinate dehydrogenase	27.16 ± 2.24	15.71 ± 1.26 ^{a*}	25.41 ± 1.32^{b*}	23.81 ± 01.80 ^{b*cNS}	28.61 ± 2.42^{dNS}
Malate dehydrogenase	370.80 ± 27.12	241.81 ± 19.21 ^{a*}	361.71 ± 21.81^{b*}	356.30 ± 23.61 ^{b*c*}	370.61 ± 27.89^{dNS}
NADH dehydrogenase	29.46 ± 2.32	15.81 ± 1.38 ^{a*}	27.96 ± 1.91^{b*}	26.91 ± 1.84 ^{b*cNS}	29.81 ± 2.23^{dNS}
Cytochrome C oxidase	6.38 ± 0.41	3.81 ± 0.28 ^{a*}	6.01 ± 0.46 ^{b*}	5.86 ± 0.87 ^{b*cNS}	6.41 ± 0.71^{dNS}

Values are expressed as mean ± S.D. for six animals. Isocitrate dehydrogenase, nmol of α-ketoglutarate liberated/(min mg protein). α-Ketoglutarate dehydrogenase, μmol of ferricyanide liberated/(min mg protein); succinate dehydrogenase, μmol of succinate oxidized (min mg protein); malate dehydrogenase, nmol of NADH oxidized (min mg protein); NADH dehydrogenase, μmol NADH oxidized (min mg protein). Cytochrome *c* oxidase, optical density × 10^{-2} min (mg protein)$^{-1}$. [a] Denotes Group I compared with Group II. [b] Denotes Group II compared with Group III and Group IV. [c] Denotes Group III compared with Group IV. [d] Denotes Group I compared with Group V. Statistical differences are expressed as $^{*}p < 0.05$. NS, not significant.

3.4.7. SA on leukemic spleen cells (TEM)

TEM was carried out on sections of spleen from control and experimental animals. Splenic cells of animals from control group revealed clearly defined white pulp showing lymphoid cells with varying degrees of development across the field, sinusoid filled leukocytes and red blood cells (RBCs). The spleen from leukemic mice showed infiltration by neoplastic cells with large nuclei exhibiting varying sizes and shapes, coarse aggregates of nuclear chromatin and dilated smooth endoplasmic reticulum (Fig. 6a). The imatinib mesylate-treated animals (Group III) show degenerative changes and apoptotic bodies in the cells of the spleen. On treatment with SA, the neoplastic cells in the spleen exhibited (Group IV) (Fig. 6b) apoptotic changes with characteristic shrinkage of the nucleus and presence of apoptotic bodies. The TEM appearance of the spleen in drug-treated animals resembled that of the control group.

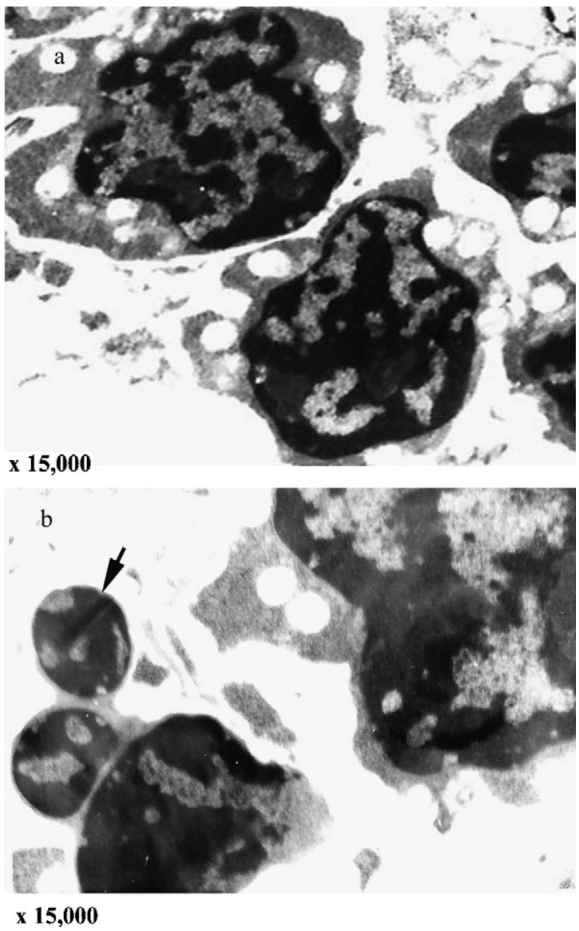

Fig. 6. Ultrastructural morphology of spleen from induced and treated Groups. (a) Actively proliferating cells from cancerous spleen and (b) apoptotic cells from SA-treated Groups.

4. Discussion

Interest in the use of herbal products has grown dramatically worldwide and a number of herbal products have been demonstrated to have anticancer activity [45]. SA is non-toxic and has antioxidant and anticancer properties [10,20,21]. It has been shown to be therapeutically effective in rats with experimental breast cancer and hepatocellular carcinoma [20,22,27]. In the present study, the antileukemic effect of SA and the restorative effect on energy metabolism in mice with experimental leukemia were evaluated. The experiments presented herein demonstrate that intravenous inoculation of 12B1 *BCR–ABL*$^{+}$ cells into BALB/c mice results in a disseminated acute leukemia of the myeloid subtype, analogous to human chronic myelogenous leukemia in blast crisis. RT-PCR analysis of the *BCR–ABL* gene in peripheral blood, spleen and liver confirmed induction of leukemia in the 12B1 cell line injected animals.

Imatinib mesylate is a tyrosine kinase inhibitor which has been shown to inhibit the *BCR–ABL*$^{+}$ fusion protein activity and has been used in the treatment of human CML and other *BCR–ABL*$^{+}$ leukemias [5] and was thus used as a standard drug in our experiments. The use of imatinib mesylate on patients has been associated with side effects such low blood counts, muscle cramps, bone pain and diarrhea. A rare, but potentially serious side effect of imatinib mesylate is liver toxicity with elevations in transaminase (SGOT and SGPT), bilirubin, creatinine, alkaline phosphatase and lactate dehydrogenase and acute liver failure [46]. SA-treated and imatinib mesylate-treated leukemic animals showed normal histology of spleen and liver and absence of leukemic cells from the bone marrow as well as marked decreased expression of *BCR–ABL* fusion gene. These results suggest that SA has definite antileukemic activity. Antileukemic activity could be a result of downregulation of the *BCR–ABL* gene which plays a direct role of pathogenesis of CML or due to an antineoplastic activity of SA by some other mechanism which leads to a decrease in tumor burden (total number of tumor cell) in the treated animals. Flavone compounds such as, pectolinarin and 5,7-dihydroxy-6,4′-dimethoxyflavone which have been shown to greatly inhibit cancer cell growth [47]. Natural products, such as dietary flavonoids, have been suggested as natural proteasome inhibitors—proteasome inhibition is an attractive approach to anticancer therapy with potential use for cancer prevention and therapeutics [48]. The flavonoids, polyphenols, carbohydrates, sterols and glycosides in SA are probably responsible for tumor regression in SA-treated animals.

The sharp drop in body weight of leukemic mice may be due to cancer cachexia. Cancer cachexia results in progressive loss of body weight, which is mainly accounted for by wasting of host body compartments such as skeletal muscle and adipose tissue. On drug treatment, the gradual increase in body weight indicates the curative property of the drug. The drug (SA) did not show any adverse effects or toxic symptoms indicating its non-toxic nature. Weight loss and tissue wasting are observed in cancer patients [49]. The weight loss implies poor prognosis and shorter survival time for cancer patients. Tessitore et al. [50] reported that the drop in body weight in cancer patients is due to decreased food intake and/or absorption, which contributes to muscle wasting in tumour cachexia. However, it is currently hypothesized that the mechanisms of cancer cachexia involve the host's production of inflammatory cytokines, which in turn orchestrate a series of complex interrelated steps that ultimately lead to a chronic state of wasting, malnourishment, and death [51].

Lipid peroxides are formed by auto-oxidation of polyunsaturated fatty acids found primarily in cell membranes. An increase in the level of lipid peroxides in tissues, therefore, reflects membrane damage [52]. In many diseases, especially cancer, membrane damage often occurs in some organ or tissue, which provokes lipid peroxidation in the membrane and accelerates the disorder in structure and function of these membranes [53]. The lack of an antioxidant defense leads to an increase in lipid peroxidation and subsequent deleterious effects [54]. More than 1.5-fold increase in lipid peroxidation in the spleen and liver of leukemia-induced group may be due to increased peroxidation of the primary substrates and polyunsaturated fatty acids [55]. Significant increase in LPO associated with various forms of carcinogenesis has been documented widely and the scavengers are known to play an important role in cancer prevention. The enhanced LPO in spleen and liver may decrease mitochondrial membrane fluidity, increase the negative surface charge distribution and alter membrane ionic permeability including proton permeability, which uncouples oxidative phosphorylation [56]. LPO in Groups III and IV, i.e. IM and SA-treated animals in the present study showed reduction towards the normal levels demonstrating the protective effect of SA.

Results of this present investigation elucidate that the drug inhibits accumulation of lipid peroxides in tissues, which lie on line with previous reports [57]. This effect can be attributed to the presence of flavonoids in the drug. Flavonoids are a group of ubiquitously distributed plant polyphenols which exhibit a wide pharmacological spectrum of effects. The inhibition of lipid peroxidation may be due to the free radical scavenging property of flavonoids, which can scavenge singlet O_2, terminating peroxides by their low redox potential [58]. They also reduce lipid peroxidation by reducing the levels of malondialdehyde and conjugated dienes [59].

In accordance with the present state of scientific knowledge, there is enough evidence to indicate that the excessive production of free radicals in the organism and the imbalance between generation of reactive oxygen species and antioxidant defenses is related to processes such as cancer and several diseases [60]. Flavonoids possess antioxidative property, based on scavenging oxygen radicals [61] and also by the Hydrogen-donating capacity of the hydroxy group in each molecule. SA treatment could possibly modulate the antioxidant system [22] that could decompose the peroxides and thereby offering a protection against lipid peroxidation.

Carbohydrate metabolism is particularly affected during proliferation of cells. Cancer cells generally have an elevated rate of glucose metabolism and an abnormal pattern of energy metabolism when compared with that of normal healthy cells [62]. The biochemical changes in the activities of various enzymes reflect the extent of metabolic alterations that occur in the cancer condition. In parallel with the rise in tumor growth rate, there is an increase in the activities of the glycolytic enzymes. To the cells suffering a hypoxic environment, the modulation of glycolysis should be an important adaptive response to help cells retain energetic metabolism and fundamental physiologic function by increasing the amount of the glycolytic enzymes and enhancing the activity of glycolysis. According to Golshani-Hebroni and Bessman [63] binding of HK to mitochondria may allow a continuous ATP flux and provide energy for phosphorylation of glucose thus, resulting in increased glycolytic rate. This, in turn might stimulate protein synthesis in proliferating cancer cells. The increased activity of glycolytic enzymes that we found in the malignant tissue is consistent with an increased activity that has been reported in a number of human cancers. Hennipman et al. [56] found decreased HK activity in tissue homogenates of normal breast as compared to breast cancer. Gudnason et al. [64] have reported the presence of HK protein in the tumor tissue but not in normal breast. The decreased glycolytic enzyme activities in SA-treated rats might be attributed to the presence of flavonoids, which play an effective role in restoring aerobic glycolysis [65]. HK can be inhibited by flavonoids which have structural resemblance between adenosine and the hydroxylated, glycosylated benzopyrene nucleus [66]. Suolinna et al. [65] also reported that inhibition of glycolysis is by the activation of flavonoid that interferes with the generation of ADP and Pi which are required for glycolysis.

The activity of gluconeogenic enzymes is inhibited significantly in tumor bearing animals. The observed reduction in activity of these enzymes in tumor bearing animals in this study has been reported by other workers and may be due to the higher lactate production of neoplastic tissues [67]. In rapidly proliferating cancer cells, there is a progressive failure of gluconeogenesis, manifested most extensively in rapidly growing tumors and is yet another reason for the marked depression or the complete absence of the activities of gluconeogenic enzymes [68]. SA is known to contain flavonoids and polyphenols which have an effective role in the restoration of aerobic glycolysis [65]. Flavonoid classes are known to competitively inhibit the transport of glucose in leukemic cell types [69].

The reduction in the activities of the Kreb's cycle enzymes, MDH, SDH, α-KGDH and ICDH in the mitochondrial fractions of leukemic mice might prove the defect in aerobic oxidation of pyruvate, which might cause the low production of ATP molecules [28]. Decreased activity of these enzymes might also be due to the alteration in the morphology and ultra structure of cancer cells [70]. Our results suggest that in the leukemic condition, there is an inactivation of mitochondrial enzymes. SA treatment protects the mitochondria from peroxidation and subsequent inactivation of enzymes. Flavonoids act on the already initiated neoplastic cell mitochondria by inhibiting tumor promotion or by penetrating the malignant cell to curb neoplasia. In structure activity studies, it was found that C 2,3-double bond, a C-keto group and 3′,4′,5′-trihydroxy B-ring are significant features of flavonoids which cause lowered activities of NADH-oxidase [71]. Lowered activities of succinic oxidase by flavonoids and other phenolic compounds have been linked to the redox activity of the B ring of these compounds. This observation is supported by a previous report on mammary carcinoma where the reduced activities of TCA cycle enzymes were replenished by SA treatment [67].

The decreased activities/levels of glycolysis and increased levels of LPO, gluconeogenic and TCA cycle enzymes observed in treated mice could be due to phenolic compounds (flavonoids) present in SA. It has been reported earlier that the SA has potential antioxidant activity and it could scavenge free radicals effectively and has also been shown to inhibit the lipid peroxidation [22]. Hydrogen-donating hydroxyl groups on the aromatic ring of phenolic compounds present in SA might be responsible for the free radical scavenging and antioxidant activity of phenolics, which correlates with earlier report [72]. Interestingly, a recent review suggests that flavonoids may not only act as conventional hydrogen-donating antioxidants but also may exert modulatory actions in cells through actions at protein kinase and lipid kinase signaling pathways [73]. The aromatic ring in phenolic acid substitutions on the B-ring in flavanoid compounds of SA could cause the highest antioxidant capacity [74]. The results of the TEM studies carried out by us suggest that there is induction of apoptosis in the leukemic cells. Earlier reports show that the accumulation of flavonoids results in up regulation of anticancer genes via mediation of the phosphorylation of p53 and alteration of bax/bcl2 ratio, leading to apoptosis [21,75]. The flavonoids in SA may modulate the bax/bcl2 ratio and thus induce the cancer cell to undergo apoptosis.

In conclusion, our results suggest that *S. anacardium* Linn. nut milk extract has antileukemic activity in mice with 12B1-induced *BCR–ABL*$^{+}$ leukemia. SA treatment could possibly modulate the antioxidant system that could decompose the peroxides and thereby offering a protection against lipid peroxidation. SA treatment diminished activity of glycolytic enzymes and modulated the gluconeogenic and TCA cycle enzymes in leukemic animals, suggesting SA may inhibit the growth of cancer cells by depriving the energy metabolism and further malignant progression through apoptosis. These positive effects can be attributed to the presence of flavonoids and other synergistic compounds in the extract.

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